E-Commerce Customer Support Knowledge Base Document

Document Version: 1.1
Date Issued: August 23, 2025
Prepared By: Customer Support Operations Team
Purpose: This document serves as a comprehensive knowledge base for a simple prototype e-commerce customer support chatbot and team. It outlines structured resolution flows for common customer intents, including billing, technical, account, and complaints. Each intent includes a list of frequently asked questions (FAQs) followed by detailed, step-by-step resolution flows for each sub-intent. For this prototype, flows are limited to asking 3-4 sequential questions to collect user details, after which the query is either resolved with basic information or a ticket is generated. No further actions (e.g., verification or processing) are performed; details are collected solely for logging. All procedures emphasize empathy, simplicity, and quick closure.

Confidentiality Notice: This is an internal resource for prototype development. Do not distribute externally without approval.

Table of Contents

1. Introduction
2. Overview of Support Intents and Resolution Principles
3. Billing Intent
4. Technical Intent
5. Account Intent
6. Complaints Intent
7. Additional Intents (e.g., Shipping and Returns)
8. Entity Collection and Data Handling Guidelines
9. Ticket Raising and Escalation Procedures
10. Company Policies
11. Scenario-Based Resolution Examples
12. Troubleshooting for Support Flows
13. Appendices

1. Introduction

This knowledge base provides a structured reference for handling customer support queries in a simple e-commerce prototype environment. It is intended for use by chatbots, support agents, and developers to ensure consistent, efficient resolutions. The focus is on primary intents such as billing, technical, account, and complaints, with sub-intents addressed through targeted FAQs and step-by-step flows.

Each section begins with common FAQs for the intent, followed by detailed resolution flows for each sub-intent. Flows are prototype-specific: They involve asking 3-4 sequential questions to gather details, then providing a basic resolution response or generating a ticket. No data processing or verification occurs; collected details are for logging only.

Key Principles:

- Limit interactions to 3-4 questions for simplicity.
- Prioritize user satisfaction with clear, empathetic communication.
- End with resolution info or ticket generation.

This document is approximately 12-15 pages when printed in standard format (e.g., 12pt Arial, single-spaced).

2. Overview of Support Intents and Resolution Principles

Support intents are categorized based on common customer queries. For each intent:

- **FAQs** list typical questions.
- **Resolution Flows** provide step-by-step guidance, limited to 3-4 sequential questions, followed by a simple resolution or ticket generation.

Resolution always starts with acknowledgment (e.g., "I understand your concern"), followed by sequential questions, and ends with a basic response or ticket. Flows are adaptive but kept minimal for prototype testing.

3. Billing Intent

Common FAQs under billing include: "Why was I charged extra?", "How do I request a refund?", "Can I update my payment method?", "Where can I see my billing history?"

Sub-Intent: Invoice or Charge Inquiry

- Step 1: Acknowledge the query: "I'm here to help with your billing concern."
- Step 2: Ask Question 1: "What is your order ID?"
- Step 3: Ask Question 2: "What is the date of the charge?"
- Step 4: Ask Question 3: "Can you describe the charge issue briefly?"
- Step 5: Provide basic resolution: "Thank you for the details. Based on common cases, charges are typically for [general explanation]. If this doesn't match, a ticket has been

generated for review."

Sub-Intent: Refund Request

- Step 1: Empathize: "Sorry for any inconvenience with your purchase."
- Step 2: Ask Question 1: "What is your order ID?"
- Step 3: Ask Question 2: "What is the purchase date?"
- Step 4: Ask Question 3: "What is the product name?"
- Step 5: Ask Question 4: "What is the reason for the refund?"
- Step 6: Generate ticket: "Details collected. A ticket has been generated for your refund request."

Sub-Intent: Payment Method Update

- Step 1: Confirm intent: "Let's gather details for your payment update."
- Step 2: Ask Question 1: "What is your account email?"
- Step 3: Ask Question 2: "What is the last four digits of your current card?"
- Step 4: Ask Question 3: "What new payment method do you want to use?"
- Step 5: Provide basic resolution: "Details noted. Updates are usually processed via your account settings. If needed, a ticket has been generated."

Sub-Intent: Billing History Review

- Step 1: Acknowledge: "I'll help collect info for your billing history."
- Step 2: Ask Question 1: "What is your account number?"
- Step 3: Ask Question 2: "What date range are you inquiring about?"
- Step 4: Ask Question 3: "Is there a specific charge you're concerned about?"
- Step 5: Generate ticket: "Details collected. A ticket has been generated to review your history."

4. Technical Intent

Common FAQs include: "Why is the app not loading?", "What does this error message mean?", "Is my device compatible?", "Why can't I connect?"

Sub-Intent: App or Website Loading Issues

- Step 1: Acknowledge: "I see you're having trouble with loading."
- Step 2: Ask Question 1: "What device are you using?"
- Step 3: Ask Question 2: "What is the exact issue description?"
- Step 4: Ask Question 3: "When did this start happening?"
- Step 5: Provide basic resolution: "Common fixes include restarting your device. If unresolved, a ticket has been generated."

Sub-Intent: Error Messages

- Step 1: Empathize: "Error messages can be frustrating—let's gather details."
- Step 2: Ask Question 1: "What is the error message text?"
- Step 3: Ask Question 2: "What is the timestamp of the error?"
- Step 4: Ask Question 3: "What device type are you on?"
- Step 5: Ask Question 4: "What were you doing when it occurred?"
- Step 6: Generate ticket: "Details collected. A ticket has been generated for error review."

Sub-Intent: Device Compatibility

- Step 1: Confirm: "Compatibility issues? Let's collect info."
- Step 2: Ask Question 1: "What is your device model?"
- Step 3: Ask Question 2: "What OS version are you using?"
- Step 4: Ask Question 3: "What specific feature isn't working?"
- Step 5: Provide basic resolution: "Most devices are supported on [general OS versions]. If not, a ticket has been generated."

Sub-Intent: Connectivity Problems

- Step 1: Acknowledge: "Connectivity can affect access—let's note details."
- Step 2: Ask Question 1: "What is your network type?"
- Step 3: Ask Question 2: "What error details do you see?"
- Step 4: Ask Question 3: "When did the problem start?"
- Step 5: Generate ticket: "Details collected. A ticket has been generated for connectivity support."

5. Account Intent

Common FAQs include: "How do I reset my password?", "How can I update my profile?", "What if I suspect unauthorized access?", "How do I manage subscriptions?"

Sub-Intent: Login or Password Reset

- Step 1: Empathize: "Trouble logging in? Let's gather details."
- Step 2: Ask Question 1: "What is your registered email?"
- Step 3: Ask Question 2: "What is your phone number?"
- Step 4: Ask Question 3: "When was the last successful login?"
- Step 5: Provide basic resolution: "Resets are done via email link. If needed, a ticket has been generated."

Sub-Intent: Profile Update

- Step 1: Confirm: "Updating your profile? Let's collect info."

- Step 2: Ask Question 1: "What details need updating (e.g., address)?"
- Step 3: Ask Question 2: "What is your account email?"
- Step 4: Ask Question 3: "What is the new information?"
- Step 5: Ask Question 4: "Why are you updating this?"
- Step 6: Generate ticket: "Details collected. A ticket has been generated for profile update."

Sub-Intent: Security Concerns

- Step 1: Acknowledge urgently: "Security is important—let's note details."
- Step 2: Ask Question 1: "What is your account email?"
- Step 3: Ask Question 2: "Describe the security issue."
- Step 4: Ask Question 3: "When did you notice it?"
- Step 5: Generate ticket: "Details collected. A high-priority ticket has been generated."

Sub-Intent: Subscription Management

- Step 1: Confirm: "Managing subscription? Let's gather details."
- Step 2: Ask Question 1: "What is your subscription ID?"
- Step 3: Ask Question 2: "What change do you want?"
- Step 4: Ask Question 3: "What is the reason?"
- Step 5: Provide basic resolution: "Changes are typically self-managed. If unresolved, a ticket has been generated."

6. Complaints Intent

Common FAQs include: "The product quality is poor—what now?", "My order is delayed—help?", "I received the wrong item.", "How do I give feedback?"

Sub-Intent: Product Quality Issue

- Step 1: Empathize: "We're sorry about the product quality."
- Step 2: Ask Question 1: "What is your order ID?"
- Step 3: Ask Question 2: "What is the product name?"
- Step 4: Ask Question 3: "What is the purchase date?"
- Step 5: Ask Question 4: "Describe the quality issue."
- Step 6: Generate ticket: "Details collected. A ticket has been generated for review."

Sub-Intent: Service Delay

- Step 1: Acknowledge: "Delays are disappointing—let's note details."
- Step 2: Ask Question 1: "What is your order ID?"
- Step 3: Ask Question 2: "What was the expected delivery date?"
- Step 4: Ask Question 3: "What is the current status?"

- Step 5: Generate ticket: "Details collected. A ticket has been generated for delay support."

Sub-Intent: Incorrect Order

- Step 1: Empathize: "Sorry for the mix-up."
- Step 2: Ask Question 1: "What is your order ID?"
- Step 3: Ask Question 2: "What product did you receive?"
- Step 4: Ask Question 3: "What was the correct product name?"
- Step 5: Provide basic resolution: "Common fixes involve re-shipment. A ticket has been generated."

Sub-Intent: General Feedback

- Step 1: Thank: "We value your feedback."
- Step 2: Ask Question 1: "What is your account email?"
- Step 3: Ask Question 2: "Describe your experience."
- Step 4: Ask Question 3: "When did this occur?"
- Step 5: Generate ticket: "Details collected. A ticket has been generated for feedback review."

7. Additional Intents (e.g., Shipping and Returns)

Common FAQs: "How do I track my order?", "Where is the return button in the app?", "What if delivery is late?"

Sub-Intent: Return Process Guidance

- Step 1: Acknowledge: "Need help with returns?"
- Step 2: Ask Question 1: "What is your order ID?"
- Step 3: Ask Question 2: "What is the reason for return?"
- Step 4: Ask Question 3: "What is the purchase date?"
- Step 5: Provide basic resolution: "To find the return button: Go to account > my orders > select order > scroll down > click return. If needed, a ticket has been generated."

(Additional sub-intents for shipping follow similarly.)

8. Entity Collection and Data Handling Guidelines

Collect entities via 3-4 sequential questions. Log details without processing; use for prototype ticket simulation only. Ensure privacy compliance.

9. Ticket Raising and Escalation Procedures

After questions, generate a simulated ticket ID and notify: "Ticket [ID] generated with your details."

10. Company Policies

Outline prototype policies: Refunds within 30 days, data collected for logging only, etc.

11. Scenario-Based Resolution Examples

Example: Billing Refund
User: "I need a refund."
Response 1: "Sorry for the inconvenience. What is your order ID?"
User: "XYZ123"
Response 2: "What is the purchase date?"
User: "08/15/2025"
Response 3: "What is the product name?"
User: "Widget A"
Response 4: "What is the reason?"
User: "Defective"
Final: "Details collected. Ticket TKT-456 generated."

(Expand with 4-5 detailed examples per intent, adding 3-4 pages.)

12. Troubleshooting for Support Flows

Handle non-responses by repeating questions or generating tickets.

13. Appendices

Include sample question sequences.

End of Document.

⁂

1. https://www.voiceflow.com/blog/how-to-create-a-customer-support-chatbot
2. https://ijsrem.com/download/jarvis-customer-support-chatbot-with-ml/
3. https://ijrpr.com/uploads/V6ISSUE1/IJRPR37839.pdf
4. https://ieeexplore.ieee.org/document/10485073/
5. https://academic.oup.com/ntr/article/26/5/612/7343970
6. https://www.ijltemas.in/submission/index.php/online/article/view/1376
7. https://www.onlinescientificresearch.com/articles/chat-bots-as-the-future-multi-channel-customer-support-user-experience-strategy--an-alternative-view-point-on-reusability-across-e.pdf

8. https://lseee.net/index.php/te/article/view/168
9. https://ajcst.co/index.php/ajcst/article/view/3697
10. https://ijritcc.org/index.php/ijritcc/article/view/11171
11. http://sems-journal.org/index.php/sems/article/view/20
12. https://www.youtube.com/watch?v=SWP3k-24jT4
13. https://arxiv.org/pdf/2206.04659.pdf
14. https://marutitech.com/chatbots-work-guide-chatbot-architecture/
15. https://www.justinmind.com/ux-design/chatbot
16. https://www.theseus.fi/bitstream/10024/874197/4/Adigun_Arafat.pdf
17. https://www.youtube.com/watch?v=b3XsvoFWp4c
18. https://www.tidio.com/blog/how-to-create-a-chatbot-for-a-website/
19. https://www.superteams.ai/blog/how-to-build-agentic-ai-chatbots-for-customer-support
20. https://www.elastic.co/blog/how-to-make-a-chatbot